Project Title

String Matching Application using the Z Algorithm

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

String matching is a fundamental problem in computer science and is widely used in applications such as text searching, information retrieval, plagiarism detection, DNA sequence analysis, and search engines. This project presents a **String Matching Application using the Z Algorithm**, an efficient pattern-matching technique that finds all occurrences of a given pattern within a text in linear time, **O(n + m)**, where *n* is the length of the text and *m* is the length of the pattern. The algorithm constructs a **Z-array**, which stores the length of the longest substring starting at each position that matches the prefix of the combined pattern and text.

The application combines the pattern and text using a special delimiter and computes the Z-array to efficiently identify pattern occurrences. Whenever a Z-value equals the length of the pattern, the corresponding position represents a successful match. The application displays all matching positions and reports **"Pattern not found"** when no occurrence exists. This project demonstrates the practical application of the Z Algorithm and highlights its efficiency in handling large strings, making it suitable for real-world applications such as document searching, plagiarism detection, and sequence analysis.

Signature of the Guide